PROFESSOR'S SUMMARY / FOUR FIXED TARGETS

Polynomial images on a Cartesian square

Can a fixed polynomial take substantially fewer values than there are input pairs? For each polynomial below we seek a constant $c_P > 0$ and arbitrarily large finite $A \subset \mathbb{R}$ such that

$$N = |A|, \qquad |P(A, A)| \leq N^{2-c_P}.$$

The same A supplies both coordinates. Every one of its N^2 ordered pairs contributes to the image. A constant-factor saving at one size cannot establish a uniform exponent.

Where the plus-sign cubic stands

T01 OPEN. For $P(x, y) = (x - y)^3 + x + y$, fixing an output identifies its *fiber*, the ordered pairs with that exact value. Algebraic curves produced finite sets with repeated outputs. On one six-point set, two outputs each occur four times, while the other 28 pairs all give different outputs. The full image has 30 values. The associated marked elliptic point has infinite order. Its orbit cannot stay within the finite set of compatible frame points. [E3–E4]

The October branching theorem gives a test that any successful family must pass. Divide each scalar interval into four at each generation, stopping once its points are separated. A pair branches when another child square in the same parent contains its exact output. If the target bound holds, then for $0 < \gamma < c_P$ all but $N^{2-\gamma}$ ordered pairs must branch more than $\lfloor (c_P - \gamma) \log_8 N \rfloor$ times. The theorem leaves open how to construct such sets. [E6]

Three other fixed targets

Track	Polynomial	Result and remaining gap
T02	$(x + 2y)^4 + 3x - y$	Exact support in a ramified local quotient. The resulting real packing estimate has no positive gap. [E7]
T03	$x^2 + xy + y^2$	Many norm-preserving units are available. A bound on typical returns of actual input pairs remains open. [E8]
T04	$(x - y)^3 - 3(x + y)$	Exponential growth of distinct real inputs is proved for one affine recursion. Its full output growth is unknown. [E9]

Evidence boundary. All four targets remain open. The report reproduces the elementary branching argument and summarizes identified project proofs elsewhere. Finite counts are inherited exact checks. The export adds no research or external review. Source keys refer to the full report's evidence appendix.

T01 / SEPTEMBER

From a small image to a shared scalar

The cost of a partially controlled square

Suppose a construction prescribes outputs on $E \subseteq A^2$. Write $Q = |E|$, $Z = |P(E)|$, $B = N^2 - Q$. Even with no information about the complement, one has

$$|P(A, A)| \leq Z + B.$$

The B uncontrolled pairs can each cost a new value. A large fiber alone only shows a saving on its own pairs. Collision energy $\sum_z |F_z|^2$, where $F_z = \{(x, y) \in A^2 : P(x, y) = z\}$, gives the *lower* bound $N^4 / \sum_z |F_z|^2$ for image size. It cannot replace the upper bound above.

16–21 September / A finite gain and its limit

Start at $A = \{0, 2\}$. The three branches $x \mapsto x/2 + j/3$ for $j = 0, 1, 2$ give six distinct scalars. Exact enumeration of their square gives 22 outputs versus four on the seed. The nine new ordered blocks therefore cost $22/(9 \cdot 4) = 11/18$ of independent ninefold growth. Some entire difference slices overlap. At the next iteration two branches give the same input, $F_0(2/3) = F_1(0)$. Subsequent continuation and return studies kept track of coincident scalars and the complete image. The one-step saving did not yield a uniform recursion. [E1]

30 September / A curve for coordinate reuse

For the fixed plus-sign cubic, put $u = x - y$. Solving $P(x, y) = z$ gives all the coordinates of the fiber:

$$x = c_+(z, u), \quad y = c_-(z, u), \qquad c_\pm(z, u) = \tfrac{1}{2}(z - u^3 \pm u).$$

Thus freely choosing u gives many pairs of a given value, but also introduces scalar inputs. To reuse one coordinate across two fibers, impose $c_s(z, u) = c_t(w, v)$ with $s, t \in \{+1, -1\}$. Equivalently

$$u^3 - v^3 - su + tv = z - w.$$

When the projective cubic is nonsingular, it has genus one. Singular specializations are classified separately. Genus counts the handles of a connected smooth compact complex curve. With an origin, a smooth genus-one curve has an elliptic group law. Its multiples can supply infinitely many solutions of a scalar-sharing equation. Each such solution still needs to be converted back to actual ordered pairs. [E2]

One archived rational example has $s = t = 1$, $z - w = 6$ and an infinite-order point on its Weierstrass model. A prefix of eight multiples gave 23 distinct scalars and 16 selected pairs; its full 529-pair square had 515 outputs. This finite calculation shows why the number of group points and the number of repeated pair values must be counted separately. [E2]

T01 / 6–11 OCTOBER

Which sharing patterns can be joined?

Gluing triangles

A directed triangle of scalar vertices prescribes $(a, b), (b, c), (c, a)$. If triangles share an edge, its two endpoints must agree *as real numbers* across the two prescriptions. For one fixed-gap family all triangles are translates. At most $2N - 3$ distinct pairs are controlled, and a translated strip attains this bound. [E3]

The variable-gap construction organizes triangle pairs in a band $1 \le |x - y| \le k$ on $N = k^3$ integers, $k \ge 3$. It controls $2kN - k(k + 1)$ pairs distributed among $2k^3 - k$ outputs. This gives order $N^{4/3}$ selected pairs, hence a vanishing fraction of the square. An increasing real deformation preserving all the designated output equalities in its 27-point example, with the first scalar fixed at zero, is forced to be the original unit-spaced set. [E3]

A four-pair frame

Place rows at $q \pm h$ and columns at $q \pm s$, where $h, s > 0$ and $h \ne s$. Subtract the outputs in the two columns of a fixed row. Each row gives one repeated output precisely when

$$3h^2 + s^2 = 1.$$

The two output values then have midpoint $2q$ and difference $C(h) = 8h(1-2h^2)$. Two frames can therefore share scalars and outputs only under simultaneous restrictions on centers, row widths and column widths.

Construction	N	Q	Z	B	$Z + B$	**Actual image**
Different-center bridge	6	8	3	28	31	29
Same-center two-width bridge	6	8	2	28	30	30

The table uses distinct scalars and ordered pairs. The first bridge has one common output on four pairs; the same-width translated ladder built from it contains an arithmetic progression whose subsquare already has at least $N^2/4$ outputs. The second bridge has two four-pair fibers and 28 singleton fibers in the full square. Its stronger local repeat is compatible with a larger actual image because the complement differs. [E3–E4]

Two different geometric obstructions

Fix ordered outputs $z > w$, $c = z - w$, and require two columns to complete the *same two rows*. The admissible positive widths solve $C(h) = c$ with $0 < h < 1/\sqrt{3}$ and $h \ne s$. At most two widths and four real marked sharing points survive. The fixed-output elliptic curve may still contain infinitely many other real points. [E4]

T01 / 11–21 OCTOBER

Completing frames and testing their rigidity

A second column and the genus-four cover

For general one-column sharing of two distinct outputs with the same second coordinate, the curve $u^3 - v^3 + u - v = c$ has the model

$$E_c : \quad Y^2 = X^3 - 48X - 128 - 432c^2.$$

Completing *one row* to another column requires a choice $u' = (-u \pm W)/2$, $W^2 = 4 - 3u^2$. Adjoining W gives a cover, meaning that generic sharing points have two completion choices. Six simple branch points (where the choices coincide) give a smooth normalization of genus four. An isogeny is a nonconstant algebraic group map between elliptic curves. Pulling this cover back under a translated isogeny of degree e gives genus $1 + 3e$. The genus formula excludes a rational choice of completion over the whole elliptic parameter curve. Individual frames are still possible. This is a different condition from fixing the same two complete rows. [E4]

One marked difference has infinite order

The same-center six-scalar bridge singles out two positive-Y points R_1, R_2. Its marked shift is $T = R_2 - R_1$. Finite checks for orders $2, 3, 4$ left arbitrary order unresolved. The subsequent proof works in a field of degree eight. Its involution σ fixes R_2 and negates R_1, giving $T + \sigma T = 2R_2$. At a specified embedding into $\mathbb{Q}_{23}$ of the subfield defining R_2, the x-coordinate of $4R_2$ has valuation -2. The formal logarithm sends this point to a nonzero additive value, proving infinite order. Torsion of T would imply torsion of $2R_2$, a contradiction. This resolves the order of that one shift. Its iterates need not be compatible frames. [E4]

Three outputs, then four

When the output labels vary, a three-output junction can reuse more slots: its eight actual scalars control ten pairs with three values. Along the proposed extension $N = 4m$, $Q = 6m - 2$ and $Z = m + 1$; one fixed-step progression gives $|P(A, A)| \geq N^2/16$. The changed single-role three-output junction was subsequently excluded by incompatible width inequalities. [E5]

A four-output junction uses the six edges of a tetrahedron as output pairs. For reflection-symmetric gaps (a, b, a), four registered role assignments contradict the real conic and output-gap equations. A fifth, separately specified assignment also fails: its hypotheses force the same signed quantity to be both below and above one. These are exact obstructions for the recorded assignments. No admissible instance exists there to which one could assign an image count. The unrestricted four-output problem remains open. [E5]

T01 / 21 OCTOBER

A necessary pattern on exact fibers

Choose a half-open interval containing A. At each generation divide each scalar interval into four equal children, giving sixteen child squares per parent. Stop at L once the scalar intervals separate the points of A. For $\xi \in A^2$, let $b_L(\xi)$ count generations where its own exact output $P(\xi)$ occurs in at least two child squares of its parent.

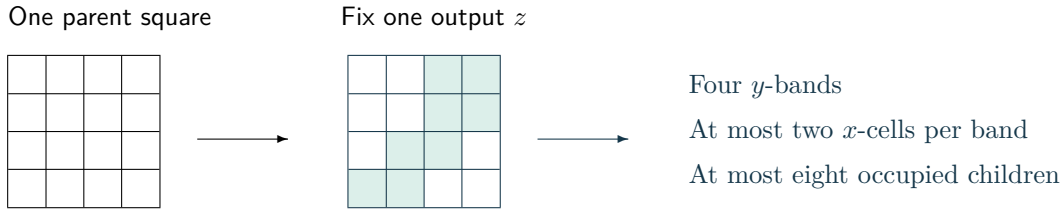
PROVED MC-S1/S2. For every integer $b \geq 0$,

$$\#\{\xi \in A^2 : b_L(\xi) \leq b\} \leq |P(A,A)|\, 8^b.$$

Proof (complete). At fixed y, $P_x = 1 + 3(x-y)^2 \geq 1$ and $P(x,y)$ tends to opposite infinities as $x \to \pm\infty$. Every real output thus has a unique graph $x = f_z(y)$. For $d = f_z(y) - y$, implicit differentiation gives

$$f_z'(y) = \frac{3d^2 - 1}{3d^2 + 1}, \qquad |f_z'(y)| \leq 1.$$

Within a half-open child y-interval the finite occupied y-values span less than the child width. The occupied x-values have no larger span and hence meet at most two child intervals. There are four y-children, so at most eight occupied child squares per exact output.



Schematic occupancy bound. Shading illustrates a capacity, not recorded observations.

Suppress the unbranched vertices of each output's occupied tree. Its leaves correspond to actual pairs, their depths equal b_L, and each internal vertex has at most eight children. Label edges to siblings by different symbols from an eight-letter alphabet. The leaf words are prefix-free. The extensions to length b of all leaves of depth at most b are nonempty and disjoint, so there are at most 8^b such leaves. Summing over the actual outputs proves the inequality. If $|P(A,A)| \leq N^{2-c}$, substituting $b = \lfloor (c-\gamma)\log_8 N \rfloor$, $0 < \gamma < c$, gives at most $N^{2-\gamma}$ exceptional pairs. □

Scope. The theorem leaves the separation between branching levels and the balance of the splits uncontrolled. The local version charges the number of pairs in a cell and reuse of equal values across cells. Metric compression, in contrast, only covers outputs by short intervals: on one frozen 12-point nonuniform set, 19 intervals of length one suffice for its 136 distinct outputs. Finite counts and metric covers do not imply an exponent. [E6]

T02 AND T04 / SEPTEMBER CHECKPOINTS

Local output counts and real input growth

T02 / The fixed quartic

For $P_4(x, y) = (x + 2y)^4 + 3x - y$, the September checkpoint counts outputs at every precision in a specific local ring. In the local cubic extension used there, $7\mathcal{O} = \mathfrak{p}^3$ and the residue field has seven elements. Take $R_m = \mathcal{O}/\mathfrak{p}^{3m}$ and the full critical residue class $E_m = (\alpha + \mathfrak{p}) \bmod \mathfrak{p}^{3m}$ with $\alpha^3 \equiv -1 \pmod{\mathfrak{p}}$. The C4 support count is

$$|P_4(E_m, E_m)| = \begin{cases} 4, & m = 1, \\ 22 \cdot 7^{3m-4}, & m \geq 2. \end{cases}$$

For $m \geq 2$ the output support consists of three full $\mathfrak{p}^3$-cosets and one full $\mathfrak{p}^4$-coset. The relative density improves once. After both input restrictions are charged, its factor stabilizes at $22/49$. The proposed real full-interval packing must also pay for the number of inputs and the widths of the output classes. The recorded certificate fails to produce a positive net saving. Other higher-precision subsets remain outside that conclusion. [E7]

T04 / The minus-sign cubic

The fourth track asks about $P_-(x, y) = (x - y)^3 - 3(x + y)$. Its two real input maps have opposite shifts and reciprocal positive slopes. Let $\epsilon = 2 + \sqrt{3}$, $a_0 = \sqrt{3}/2$ and set

$$A_0 = \{0\}, \qquad A_{m+1} = (\epsilon A_m + a_0) \cup (\epsilon^{-1} A_m - a_0).$$

The 2^m histories need not produce different scalars. The input theorem counts injective subfamilies and controls the number of distinct affine maps. For $\Lambda > 1$ satisfying $\Lambda^{-1} + \Lambda^{-3} + \Lambda^{-5} = 1$, and $\varphi = (1 + \sqrt{5})/2$, it gives

$$|A_m| = \exp(a_* m + o(m)), \qquad \log \Lambda \leq a_* \leq \log \varphi.$$

All scalars lie in a fixed quadratic field with bounded denominator. Mixed branch pairs still move both the sum and the difference of the endpoints. The full output support has only the certified upper bound $\exp(2a_* m + o(m))$. The true exponential gap is unknown. No T01 plus-sign result supplies this estimate. [E9]

Evidence status. The two results above are reported from identified checkpoint proofs. Both are summarized here. The dated status notes distinguish recovered copies, corrections and still-missing earlier provenance.

T03 AND SHARED WORK / SEPTEMBER–NOVEMBER

Norm symmetry and the cost of a return

T03 / The Eisenstein quadratic

For $Q(x,y) = x^2 + xy + y^2$, write $\omega^2 + \omega + 1 = 0$. The product $(x - \omega y)(x - \omega^2 y) = Q(x,y)$ identifies the output with a norm. Multiplying $z = x - \omega y$ by an element of norm one keeps the norm unchanged. It gives another pair only if both transformed coordinates belong to the original real input set.

The September adaptive construction produced the exact finite $(N, |Q(A,A)|)$ values $(2,2)$, $(4,5)$, $(8,17)$, $(16,68)$, $(32,269)$. An all-level signed-root support bound, RC15/4, would give a uniform saving under its hypotheses. It remains open. Subsequent modular-lift and cross-block studies addressed the cost of realizing whole Cartesian products. [E8]

In October the balanced-factor construction reached a decisive obstruction. For that unchanged sequence of carriers A_r there are fixed $M \geq 1$ and r_* such that at all $r \geq r_*$,

$$|A_r|^2/(18M) \leq |Q(A_r, A_r)| \leq |A_r|^2.$$

This bounds the actual image, ruling out a fixed exponent saving for that family. The proof restricts multiplicity on a positive proportion of sources, including alternatives outside the selected moves. [E8]

A separate mixed-signature lattice carrier A_j uses allowed coefficients inside fixed bounds at the archimedean embeddings. On the recorded tower with level parameter d_j, one radius T_* supplies at least 2^{d_j} norm-one order units per level. On the $M_j = |A_j|^2 - 1$ nonzero source pairs define $R_j(p)$ as the number of actual pairs in the same unit orbit as p. Each occupied orbit contributes one to

$$\nu_j = \sum_p \frac{1}{R_j(p)} = M_j \, \mathbb{E}_p[1/R_j(p)] \,, \qquad |Q(A_j, A_j)| \leq 1 + \nu_j.$$

An upper bound decaying exponentially in d_j for the inverse moment would give a power saving. It remains open. Counting available units does not tell how often both transformed coordinates fall back inside the set. [E8]

4 November / A separate proved family

The project reports Theorem 1 for every fixed $a, b, c, e \in \mathbb{Z}$. The polynomial $(ax + by)^2 + cx + ey$ admits a fixed power saving on arbitrarily large finite real squares. The condition $ab(ae - bc) \neq 0$ singles out nonspecial examples. The theorem includes degenerate cases. The proof imports a field-supply theorem (Proposition 3.8 in its source). The audit separates that input from internal deductions MP-01–MP-07. The quadratic part of this family has rank at most one. T03 has rank two. All four targets remain open. [E10]

EVIDENCE APPENDIX / I

How to find the arguments

The keys below identify the native sources. For E1–E6, directory names are relative to `research/notes/`. Locators name a proof by identifier or section title. The present export checked source correspondence and typesetting. Only the proof of MC-S1/S2 on page 5 is reproduced in full.

E1 / Three branches and returns

`2026-09-16-three-branch-lift/README.md`, §1 (maps, $4 \to 22$, literal iteration); `2026-09-16-three-branch-continuation/README.md` (bounded continuation). See `2026-09-21-growing-step-cubic/structure-and-reuse.md`, GS-P3, for selected coverage of the output–difference grid. Exact portable counts are in `examples/2026-09-16-three-branch-lift.json` from the repository root.

E2 / Sharing geometry

`2026-09-30-elliptic-geometry/genus.md`, EG-G1 and EG-G2, gives the fiber parameter and the sharing classification. `complex-extensions.md`, EG-C1/C2 in the same directory, gives the Weierstrass transform and infinite-order point. Its `README.md`, sharing-prefix table, records the 23-scalar example. See also the dated `verification.md`. [1]

E3 / Gluing and the first frame bridge

`2026-10-06-sharing-compatibility/construction.md`, SH-C1–C3, and `2026-10-06-variable-gap-sharing/construction.md`, the band construction and coverage ledger, give the two network counts. `2026-10-06-changed-collision-pattern/rigidity.md`, CP-R1, is the fixed-incidence rigidity proof. In the same directory, `construction.md`, CP-C1–C4, proves the conic identity, six-scalar bridge and ladder bound. `exact-audit.md` records the full-square data.

E4 / Admissible points and arithmetic

`2026-10-11-torsion-compatible-sharing/geometry.md`, TC-G1–G6, separates the finite completed-frame locus from the genus-four cover for a general row. Its `exact-results.json` and `exact_audit.py` retain the two six-point squares and bounded torsion checks. `2026-10-11-marked-shift-certificate/arithmetic.md`, MS-C1, resolves the marked shift at all orders. `referee-audit.md` records the historical internal review. The proof uses one specified good place over 23. [2]

E5 / Varying outputs and excluded closures

`2026-10-11-varying-output-frames/compatibility.md`, VO-C1–C5; `2026-10-11-vo-o1-freeze/README.md`, §3, VO-C6; `2026-10-20-tetrahedral-junction/compatibility.md`, TJ-C1–C4, with `catalogue.md` for the four roles; `2026-10-21-tj-o1-freeze/README.md`, *Complete contradiction*, TJ-C6, with `proof-check.md` for the internal check.

EVIDENCE APPENDIX / II

Status, data and evidence boundaries

E6 / Exact branching

`research/notes/2026-10-21-multiscale-collisions/`: `specialist.md`, MC-S1–S3; `README.md`, §§1–3; `experiment-summary.json` and `referee-counts.json`. Six inherited complete-square checks compare 533 ordered-pair outputs. The archival build ran no research grid. The local source for the proposed Raz–Zahl Theorem 1.18 comparison is marked **MISSING SOURCE**. No exponent from that comparison enters MC-S1/S2. [3]

E7–E10 / Track-specific sources

E7. `research/tracks/02-quartic-hybrid/STATUS.md`, *Accepted C2–C4 results* and *Carrier and signed rate*, specifies the local ring, counted set and full-interval packing obstruction. The proof is reported from the accepted C4 checkpoint. The status file identifies its local provenance.

E8. `research/tracks/03-eisenstein-cartesianization/STATUS.md`, *PROVED adaptive construction and exact finite ledger* and *OPEN selected obligation and CONDITIONAL consequence*. Under `research/notes/`, see `2026-10-11-bff-o1/README.md`, §§1–5; `2026-10-21-relative-unit-population/arithmetic.md`, MSU-A1–A3; and `2026-10-21-typical-unit-returns/README.md`, equations (11)–(15), with `incidences.md` for the orbit partition.

E9. `research/tracks/04-cubic-slices/STATUS.md`, *PROVED input theorem; OPEN full output* and *PROVED exact pair evolution; missing output count*. Its named input proof and audit are identified local records. A depth-30 original log remains a **MISSING PROVENANCE FILE**.

E10. `docs/release/2026-11-05/shared-quadratic.md`, Theorem 1, MP-01–MP-07, gives the public mathematical record of the 4 November audit. The original archive manifest and source-conflict register are `research/manifest.json` and `research/MIGRATION_CONFLICTS.md`. Certain underlying local reference PDFs, administrative records and large raw outputs are retained separately. Their exclusion from a public export is distinct from missing provenance.

Reading the labels. PROVED means a complete argument is identified in the source system. CONDITIONAL conclusions require the named hypotheses. FAILED describes only the specified attempt. EXPERIMENTAL counts have finite range. OPEN identifies a statement without resolution in the record. Historical internal checks are attributed at their original dates. This export certifies no external review or novelty.

Selected native reports

[1] Elliptic geometry for the cubic image problem, 2026. Internal project checkpoint, 30 September. See evidence key E2.

[2] Torsion compatible sharing and the finite frame obstruction, 2026. Internal project checkpoint, 11 October. See evidence key E4.

[3] Actual cubic collisions across scales, 2026. Internal project checkpoint, 21 October. See evidence key E6.